RFQ Automation Platform - Complete Feature Descriptions

Authentication & Identity Features

Admin Login & Management

- **Admin Authentication**: Secure login system for platform administrators using JWT tokens
- **Admin User Management**: Create, edit, and manage platform-level admin accounts
- **Role-based Access**: Support for 'admin' and 'super_admin' roles with different permission levels
- **Session Management**: JWT-based authentication with refresh tokens for security

Company Authentication (Frontend+ backend)

- **Company Login**: Multi-tenant login system where company users access their isolated data
- **User Profile Management**: Fetch and update current user profile information
- **Password Security**: Bcrypt password hashing with salt rounds for secure storage

Admin Dashboard Features

Platform Management

- **Global Company Overview**: View all companies registered on the platform
- **Company Account Management**: Create new company accounts, edit existing ones
- **Company Status Control**: Activate or deactivate company accounts to control access
- **Admin Account Management**: Full CRUD operations for admin users

Analytics & Monitoring

- **Global Platform Analytics**: View system-wide metrics, usage statistics, and performance data
- **Email Usage Metrics**: Monitor total emails sent across all companies, delivery rates

- **Subscription Overview**: Track billing status, plan distribution, and revenue metrics
- **System Health Monitoring**: Monitor platform performance and usage patterns

Support System

- **Ticket Management**: Handle support requests from companies
- **System Feature Management**: Configure available features for different subscription tiers

Company Management Features

User Management (Frontend+ backend)

- **Company User Directory**: List all users within a company with their roles and status
- **User Creation**: Add new team members with specific role assignments
- **Role Management**: Assign and modify user roles (Admin, Manager, Employee)
- **User Status Control**: Activate, deactivate, or remove users from the company
- **Permission Control**: Role-based access control determining feature availability

Team Collaboration

- **Role-based Permissions**:
    - **Admin**: Full company access, user management, billing
    - **Manager**: RFQ management, analytics, team oversight
    - **Employee**: RFQ operations, contact management (limited access)

Shipping Line Management (Frontend+ backend)

Carrier Database

- **Shipping Line Registry**: Maintain a database of shipping carriers (Maersk, MSC, CMA CGM, etc.)
- **Carrier Information**: Store carrier details including SCAC codes, headquarters, websites
- **Carrier Status Management**: Archive inactive carriers while preserving historical data

- **Custom Carrier Addition**: Add regional or specialized carriers not in standard databases

Carrier Organization

- **Hierarchical Structure**: Organize carriers by trade lanes, services, or company preferences
- **Carrier Notes**: Add internal notes and observations about each carrier's performance
- **Carrier Categorization**: Tag carriers by specialization, reliability, or service quality

Contact Management (Frontend+ backend)

Contact Database

- **Contact Directory**: Comprehensive database of contacts within each shipping line
- **Contact Information**: Store names, emails, phone numbers, job titles, departments
- **Contact Relationships**: Link contacts to their respective shipping lines and roles

Contact Intelligence

- **Performance Scoring**: AI-calculated performance scores (0-100) based on response rates, quote quality
- **Response Tracking**: Monitor how quickly and frequently contacts respond to RFQs
- **Contact Notes**: Add personal notes, preferences, and communication history
- **Tagging System**: Categorize contacts by specialization, seniority, or reliability

Contact Management Tools

- **Bulk Import**: CSV import functionality for adding multiple contacts at once
- **Do Not Contact**: Flag contacts who should not receive communications
- **Primary Contact**: Designate main contacts for each shipping line
- **Archive Management**: Soft delete contacts while preserving historical data

RFQ (Request for Quote) Management (Frontend+ backend)

RFQ Lifecycle Management

- **RFQ Creation**: Create detailed shipping quote requests with comprehensive cargo information
- **Draft Management**: Save and edit RFQs before sending to contacts
- **RFQ Details**: Include origin/destination ports, cargo details, container types, timelines
- **Status Tracking**: Monitor RFQ status through workflow (Draft → Sent → Closed → Awarded)

RFQ Information Management

- **Cargo Specifications**: Container types (20GP, 40GP, 40HC), quantities, weight, volume
- **Shipping Terms**: Incoterms, cargo ready dates, quote deadlines, urgency levels
- **Special Requirements**: Additional services like customs clearance, insurance, door delivery
- **Route Information**: Origin and destination ports with trade lane identification

RFQ Operations

- **RFQ Distribution**: Send RFQs to selected contacts with personalized messaging
- **RFQ Tracking**: Monitor which contacts received, opened, and responded to RFQs
- **RFQ Analytics**: Track response rates, average quotes, and performance metrics per RFQ
- **RFQ Closure**: Close RFQs when quotes are received and evaluated

Template Management (Frontend+ backend)

Template System

- **Template Library**: Save frequently used RFQ configurations as reusable templates
- **Template Creation**: Create templates from existing RFQs to standardize processes
- **Template Customization**: Modify templates for different trade lanes or cargo types
- **Default Templates**: Set default templates for common shipping routes

Template Features

- **Dynamic Fields**: Templates with placeholders for variable information
- **Multi-language Support**: Templates in different languages for international operations
- **Template Categorization**: Organize templates by route, cargo type, or frequency

Email Send Engine (Frontend+ backend)

Bulk Email System

- **Mass Distribution**: Send RFQs to multiple contacts simultaneously with rate limiting
- **Email Personalization**: Customize emails with recipient names, company details, RFQ specifics
- **Template Integration**: Use predefined templates with dynamic content insertion
- **Delivery Management**: Queue emails to avoid overwhelming recipients or triggering spam filters

Email Tracking

- **Delivery Tracking**: Monitor email delivery status (Queued, Sent, Delivered, Bounced, Failed)
- **Engagement Tracking**: Track email opens, link clicks, and recipient engagement
- **Bounce Management**: Handle bounce-backs and update contact status accordingly
- **Performance Analytics**: Analyze email campaign performance and delivery rates

Email Customization

- **Custom Templates**: Create company-specific email templates with branding
- **Multi-language Support**: Send emails in recipient's preferred language
- **Company Branding**: Include company logos, signatures, and custom footers
- **Scheduled Sending**: Schedule emails for optimal delivery times

Automated Follow-Up Engine

Follow-Up Automation

- **Rule-Based Follow-ups**: Automatically send follow-up emails based on configurable rules
- **Conditional Logic**: Send follow-ups only if emails weren't opened or no reply received
- **Follow-Up Scheduling**: Set custom delays (e.g., 3 days, 1 week) between follow-ups
- **Maximum Limits**: Set maximum number of follow-ups to avoid over-communication

Follow-Up Management (Frontend+ backend)

- **Follow-Up Templates**: Create specific templates for first, second, third follow-ups
- **Escalation Rules**: Increase urgency or change messaging in subsequent follow-ups
- **Follow-Up Tracking**: Monitor which follow-ups were sent and their effectiveness
- **Manual Override**: Manually skip or send follow-ups as needed

Smart Follow-Up Logic

- **Response Detection**: Automatically stop follow-ups when replies are received
- **Engagement-Based**: Modify follow-up frequency based on contact engagement history
- **Contact Preferences**: Respect individual contact communication preferences

Reply Ingestion & AI Parsing (Frontend+ backend)

Email Integration

- **IMAP Integration**: Connect to email accounts to automatically read incoming replies
- **Webhook Support**: Receive emails through webhook integration with email providers
- **Thread Matching**: Automatically match replies to original RFQs using message threads

AI-Powered Parsing

- **Price Extraction**: Automatically extract pricing information from email text
- **Currency Detection**: Identify currencies and convert to standard formats
- **Terms Extraction**: Parse validity dates, payment terms, and shipping conditions
- **Confidence Scoring**: Provide confidence levels for AI-extracted information

Data Processing

- **Manual Review**: Flag low-confidence extractions for manual verification
- **Data Validation**: Validate extracted information against business rules
- **Historical Learning**: Improve parsing accuracy based on manual corrections
- **Multi-format Support**: Parse quotes from various email formats and attachments

Quote Management (Frontend+ backend)

Quote Analysis

- **Quote Comparison**: Side-by-side comparison of all quotes received for an RFQ
- **Price Analysis**: Identify best offers, average prices, and price ranges
- **Terms Comparison**: Compare validity dates, payment terms, and additional charges
- **Carrier Comparison**: Analyze quotes by shipping line performance and reliability

Quote Processing

- **Manual Entry**: Manually input quotes received via phone, WhatsApp, or other channels
- **Quote Validation**: Verify quote completeness and accuracy
- **Quote Status**: Track quote validity, expiration, and award status
- **Winner Selection**: Mark winning quotes and notify successful carriers

Quote Intelligence

- **Historical Analysis**: Compare current quotes with historical pricing for same routes
- **Market Trends**: Identify pricing trends and market conditions

- **Carrier Benchmarking**: Compare carrier pricing and service levels
- **Rate Optimization**: Identify opportunities for better pricing

Analytics & Reporting (Frontend+ backend)

RFQ Analytics

- **RFQ Performance**: Track response rates, quote counts, and conversion rates per RFQ
- **Response Time Analysis**: Monitor how quickly carriers respond to RFQs
- **Quote Distribution**: Analyze quote price distributions and competitive landscapes
- **Success Metrics**: Track RFQ-to-booking conversion rates and award percentages

Carrier Performance Analytics

- **Response Rate Tracking**: Monitor which carriers consistently respond to RFQs
- **Quote Quality Analysis**: Evaluate quote competitiveness and accuracy
- **Win Rate Analysis**: Track which carriers win the most awards
- **Relationship Metrics**: Assess carrier relationship strength and reliability

Business Intelligence

- **Route Performance**: Analyze performance by trade lanes and shipping routes
- **Historical Benchmarking**: Compare current performance with historical data
- **Cost Analysis**: Track shipping cost trends and optimization opportunities
- **Market Intelligence**: Identify market conditions and pricing trends

Reporting Tools

- **Dashboard Views**: Real-time dashboards with key performance indicators
- **Export Functionality**: Export reports to CSV, PDF, or Excel formats
- **Scheduled Reports**: Automated report generation and distribution
- **Custom Reports**: Create custom reports based on specific business needs

Company & Billing Settings (Frontend+ backend)

Company Profile Management

- **Company Information**: Manage company details, contact information, and branding
- **User Preferences**: Set default settings for RFQ creation and email templates
- **Communication Settings**: Configure email signatures, follow-up rules, and notification preferences
- **Integration Settings**: Manage API keys, webhook endpoints, and third-party integrations

Subscription Management

- **Plan Overview**: View current subscription plan details and included features
- **Usage Tracking**: Monitor usage against plan limits (RFQs, emails, contacts)
- **Billing History**: Access invoice history and payment records
- **Plan Management**: Upgrade, downgrade, or modify subscription plans

Billing Integration

- **Stripe Integration**: Secure payment processing through Stripe
- **Payment Methods**: Manage credit cards and payment preferences
- **Automated Billing**: Automatic monthly/yearly billing with usage overages
- **Cost Optimization**: Recommendations for plan optimization based on usage patterns

Integrations & API

RESTful API

- **Comprehensive API**: Full API coverage for all platform features
- **Authentication**: Secure API access with key-based authentication
- **Rate Limiting**: API rate limiting to ensure fair usage and system stability
- **Documentation**: Complete API documentation with examples and use cases

Webhook System

- **Outgoing Webhooks**: Send real-time notifications for events (quote received, RFQ sent)
- **Incoming Webhooks**: Receive RFQ creation requests from TMS/ERP systems
- **Event Management**: Configure which events trigger webhook notifications

- **Webhook Security**: HMAC signature verification for webhook authenticity

Third-Party Integration

- **TMS/ERP Integration**: Connect with transportation management and ERP systems
- **Email Provider Integration**: Support for SendGrid, Mailgun, and other email services
- **Calendar Integration**: Sync important dates with calendar systems
- **CRM Integration**: Export contact and opportunity data to CRM systems

Infrastructure & Security

Multi-Tenant Architecture

- **Data Isolation**: Complete separation of data between companies
- **Scalable Design**: Architecture designed to support thousands of companies
- **Resource Optimization**: Efficient resource usage across multiple tenants
- **Performance Isolation**: Ensure one company's usage doesn't affect others

Security Features

- **Role-Based Access Control**: Granular permissions based on user roles
- **HTTPS Encryption**: All communications encrypted with SSL/TLS
- **Data Encryption**: Sensitive data encrypted at rest and in transit
- **Audit Trails**: Comprehensive logging of all user actions and system events

Email Security

- **SPF/DKIM Authentication**: Email authentication to improve deliverability
- **Reputation Management**: Monitor and maintain email sending reputation
- **Bounce Handling**: Automatic handling of bounced emails and list hygiene
- **Anti-Spam Compliance**: Compliance with anti-spam regulations and best practices

File Management

- **Secure File Storage**: Encrypted file storage for attachments and documents
- **Access Control**: Role-based access to uploaded files and documents
- **Virus Scanning**: Automatic scanning of uploaded files for security threats

- **Backup & Recovery**: Regular backups and disaster recovery procedures